



Research paper

The barriers to widespread adoption of vehicle-to-grid: A comprehensive review

Andrei Goncearuc^{a,b,*}, Cedric De Cauwer^a, Nikolaos Sapountzoglou^a, Gilles Van Kriekinge^a, Dominik Huber^a, Maarten Messagie^a, Thierry Coosemans^a

^a ETEC Department & MOBI Research Group, Vrije Universiteit Brussel (VUB), Pleinlaan 2, 1050 Brussel, Belgium

^b BUTO Department & MOBI Research Group, Vrije Universiteit Brussel (VUB), Pleinlaan 2, 1050 Brussel, Belgium

ARTICLE INFO

Keywords:

Vehicle-to-grid (V2G)

Electric Vehicle (EV)

Barriers

Risk matrix

Cross-impact analysis

ABSTRACT

The remarkable growth of electric vehicles (EV) popularity in recent years has significantly pushed the EV charging demand, potentially causing significant pressure on the electricity grid. The Vehicle-to-Grid (V2G) technology, allowing for bidirectional energy flow between the EV battery and the grid, is able to mitigate this pressure, by unlocking the energy flexibility, inherently available in the EV battery. However, despite its potential, V2G technology has not yet reached the mass market due to a number of barriers hampering its adoption. The current study has identified and analyzed 23 barriers potentially hampering the adoption of V2G technology, including technical, business, and EV user-related challenges. The results of the performed analysis define the risk of non-adoption of V2G technology associated with every barrier, along with the interrelations between the barriers. These research findings allow to understand the relative importance of the barriers, structuring and facilitating the necessary efforts of the involved stakeholders toward the adoption of V2G technology.

1. Introduction

The adoption of electric vehicles (EVs) has significantly accelerated in recent years (IEA, 2024), despite the fundamentally different underlying powertrain technology compared to the currently dominant internal combustion engine (ICE) technology (Wu et al., 2015). Range anxiety, high purchase cost, and lack of charging infrastructure are typically stated as the main challenges to the mass adoption of EVs (Haddadian et al., 2015). However, as a result of large investments in charging networks, encouraged by European guidelines, national action plans, and private investments of small and medium enterprises (SMEs) and energy companies, these challenges have been (at least partially) mitigated for a larger segment of the mobility fleet. The ambitious CO₂ targets imposed by the European Union (EU) result in national action plans, pushing the EV adoption towards the early majority in the next couple of years and late majority in about 5–10 years (European Council, 2024a). With the major transition expected in the next 5 years, the EU expects to accommodate between 33–40 million EVs with an annual growth of 4 million EVs, and recommends 1.3 million and 3 million public charging points to be installed by 2025 and 2030 respectively, fulfilling the new charging requirements (European Council, 2024b). The recommended charging solutions need to: (a) cover more (all) types of EV users with different

requirements with regards to e.g. power levels, location, and cost structures, (b) deal with the impacts on the energy system, and (c) be profitable for involved businesses, and cost-competitive for the EV users (compared to ICE vehicle costs per kilometer) (Anon, 2021a).

The main way to address these requirements is to innovate and develop cost-effective and “smart” charging systems, which would have to be able to unlock the energy flexibility inherently available in the battery of the EV. This flexibility can be created by the technological opportunity to adjust the timing power level, and direction of energy flow, while connected to a charger. The technology, extending the “smart” charging functionality and allowing to perform the all above-mentioned tasks already exists and is called Vehicle-to-grid (V2G) technology (Kempton and Tomić, 2005). Literature (Kempton and Tomić, 2005; Geske and Schumann, 2018; Yilmaz and Krein, 2012; Mwasilu et al., 2014; Ma et al., 2012) has demonstrated that the application of V2G technology can contribute to peak shaving, self-consumption, imbalance support (frequency control), reactive power compensation, and when introduced in mass scale, would allow the increase of the share of renewables in the energy mix. Elia, the Belgian Transmission System Operator (TSO) has acknowledged the potential of EVs for power system services and published their vision of using

* Corresponding author at: ETEC Department & MOBI Research Group, Vrije Universiteit Brussel (VUB), Pleinlaan 2, 1050 Brussel, Belgium.

E-mail address: goncearuc.andrei@vub.be (A. Goncearuc).

List of abbreviations

AC	Alternating current
BSPs	Balancing Service Providers
CCS	Combined Charging System
CPOs	Charge Point Operators
CHAdemo	CHARge de MOve
DC	Direct current
DSO	Distribution System Operator
EMS	Energy management system
ENTSO-E	European Network of Transmission System Operators for Electricity
EU	European Union
EV	Electric Vehicle
EVSE	Electric vehicle supply equipment
ICE	Internal Combustion Engine
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
MDI	Matrix of Direct Influence
MICMAC	Matrice d'Impacts Croiss Multiplication Appliqus  un Classement
MSPs	Mobility Service Providers
OCHP	Open Clearing House Protocol
OCPI	Open Charge Point Interface
OCPP	Open Charge Point Protocol
OICP	Open Intercharge Protocol
RES	Renewable Energy Sources
SMEs	Small and Medium Enterprises
SOC	State-of-charge
TSO	Transmission System Operator
V2G	Vehicle-to-grid
V2A	Vehicle-to-Aids
V2B	Vehicle-to-Building
V2C	Vehicle-to-Community
V2H	Vehicle-to-Home
V2L	Vehicle-to-Load
V2MG	Vehicle-to-Microgrid
V2N	Vehicle-to-Neighbor
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything

the EVs to move from passive consumers to prosumers, actively participating in energy and balancing markets (Elia, 2020a). The research has shown that large-scale adoption of EVs, powered by the V2G technology, in Belgium by 2030 can lead to the doubling of the installed capacity of wind (to 15 GW), reduce the critical excess energy and curtailment with 3 TWh, and, as a consequence reduce the greenhouse gas emissions of the energy production and mobility by 30%. However, while smart chargers are already becoming a common market practice, V2G, despite its potential, is still very much in the pilot project or prototype stage and has large difficulties getting market traction.

The first steps towards the identification of the challenges hampering the adoption of V2G technology have already been made by recent studies. According to Sovacool et al. (2017), these challenges can be divided into four groups: technical, financial, social-environmental and behavioral. The follow-up research (Noel et al., 2019a) further specifies the existing challenges by defining a list of V2G barriers for the Nordic region, based on the interviews with the experts in the transportation and electricity fields. This study has defined nine V2G barriers, including uncertainty and skepticism, lack of necessity, preference for other

technologies, insufficient EV volume, electricity market structure, poor business case/model, increased cost/complexity, battery degradation, and customer resistance. Moreover, a number of studies arise additional issues, such as security and privacy challenges (Saxena et al., 2017), the complexity of energy management system (EMS) optimization (Yi et al., 2020), the absence of V2G-specific taxation mechanism (PwC, 2019), and others.

1.1. Contributions

The contribution of the current study is twofold:

- First, considering the relative novelty of the V2G technology and the ongoing evolution of the EV charging market, the existing literature still lacks an overarching and detailed overview of the barriers, that the V2G technology has to overcome in order to get market traction. Therefore, by employing the knowledge available in the currently existing literature and internal expertise, this research identifies and defines a detailed list of the current barriers to the adoption of V2G technology (Section 4.1). Furthermore, the defined barriers are validated by the survey of the EV charging business ecosystem stakeholders' expert opinions.
- Second, according to Zhang and Fan (2010), a business ecosystem is a dynamic system of interrelated business models of its participants (e.g. suppliers, distribution channels, direct customers, governmental agencies, etc.), while the barriers to the adoption of existing technology are tightly related to its business ecosystem stakeholders (Scaringella and Radziwon, 2018). Thus, the barriers to the adoption of V2G technology are interrelated with each other as well and the solution of one barrier can potentially facilitate the solution of the others. These interrelations between the barriers create an additional attention point still missing in the existing literature. Therefore, by quantifying the qualitative expert opinions and making use of the combination of Risk Matrix (Garvey P.R., 1998; Bao et al., 2021) and Matrice d'Impacts Croiss Multiplication Appliqus  un Classement (MICMAC) (Duperrin and Godet, 1973) methods (Sections 3.1 and 3.2), the current research analyzes the potential risk levels and interrelations of all the identified barriers. The analysis results provide the understanding of the relative importance of the barriers, structuring and facilitating the necessary efforts of the involved stakeholders towards the adoption of V2G technology (Section 4.2).

2. Concepts and potential benefits of vehicle-to-grid technology

The term Vehicle-to-grid (V2G) was introduced by the research of Kempton et al. in 2001 (Kempton and Tomi, 2005) to describe the concept of bidirectional energy transfer between the EV battery and the electrical grid, schematically represented in Fig. 1. As demonstrated by green arrows in Fig. 1, the energy flows from the electricity-generating units through the electrical grid to the EV charger, representing the ordinary charging process of the EV battery. However, as indicated by two-sided arrows between the V2G-capable EV, V2G charger, and electrical grid, the V2G technology additionally allows to discharge energy from the EV battery back to the grid and use it for other purposes.

Moreover, the applications of V2G technology are not limited by the interaction of EVs with the electrical grid, expanding to the use of EV battery as a source for providing energy to an element outside the system of the vehicle (Saldaa et al., 2019; Pearre and Ribberink, 2019; Thompson and Perez, 2020). As demonstrated in Fig. 2, the separate concepts were also defined to describe the specific applications of the technology. In order to bring clarity to the nomenclature linked to V2G, the following list synthesizes and defines different V2G-related concepts shown in Fig. 2:

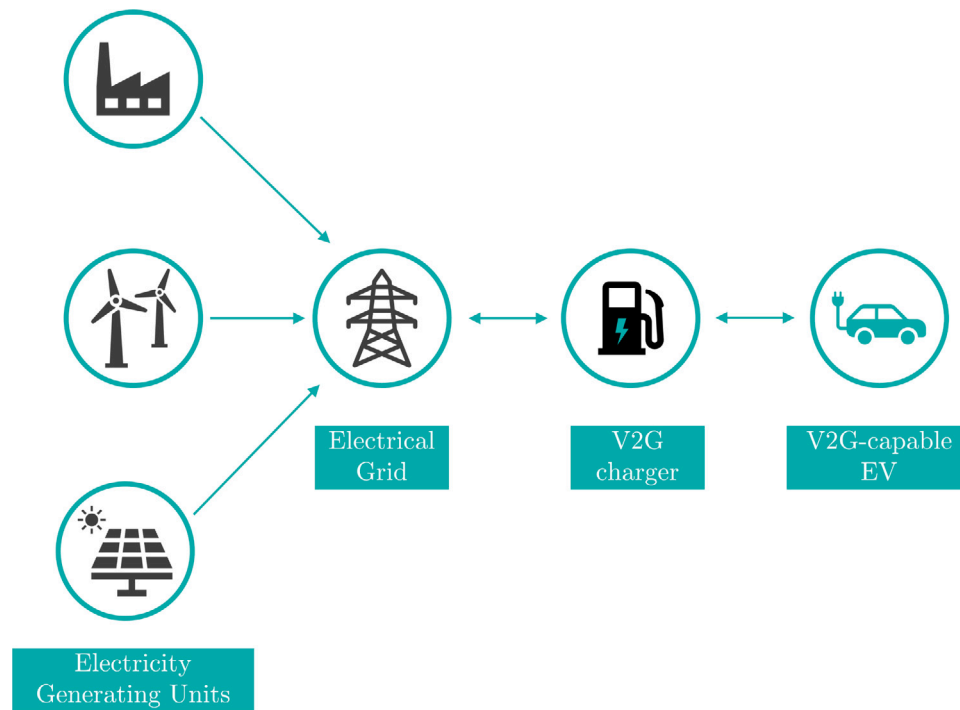


Fig. 1. Schematic representation of V2G technology.

- *Vehicle-to-Microgrid (V2MG)* - a setup where EVs act as mobile batteries that can both draw electricity from the microgrid and discharge stored energy back in case of necessity (Lu et al., 2021; Rahman et al., 2020).
- *Vehicle-to-Building (V2B) (Vehicle-to-Home, V2H)* - a concept focusing on the provision of energy available in EV battery to building/home (Ioakimidis et al., 2018; Kuang et al., 2017; Saldaña et al., 2019).
- *Vehicle-to-Aids (V2A)* - the use of energy stored in EV batteries for medical and emergency purposes (patrols, fire trucks, etc.) (Licea, 2020).
- *Vehicle-to-Vehicle (V2V)* - the use of energy stored in an EV battery to feed another EV.
- *Vehicle-to-Community (V2C)* - the aggregation of EV batteries for local use in a residential community (Noel et al., 2019b; Yamagata et al., 2014).
- *Vehicle-to-Neighbor (V2N)* - refers to energy transition from one's EV to one's neighbor (Saldaña et al., 2019).
- *Vehicle-to-Load (V2L)* - the use of an EV battery as an alternative energy source in areas with limited grid connections or for backup energy provision for various appliances (Thompson and Perez, 2020; Rodríguez-Licea et al., 2019). Some authors mention buildings, homes, vehicles and other various appliances as specific cases of the "load" in V2L and as a result classify V2B, V2H, V2V, V2N and V2A as a part of the V2L concept as well (Pearre and Ribberink, 2019).
- *Vehicle-to-Everything (V2X)* is an umbrella term (Thompson and Perez, 2020), defining the purpose of the application of V2G technology to any element outside the system of the vehicle (Pearre and Ribberink, 2019). Different abbreviations (Vehicle-to-everything, V2E), of the V2X concept have also been observed in the literature.

Along with the above-mentioned applications, the V2G-enabled bidirectional interaction of EV battery with the electrical grid allows to participate in the following energy-related services:

- *Load shifting* is the concept of shifting power consumption to avoid large demand peaks. The V2G technology allows to reduce

the existing peak demands by using EV batteries as a power supply source and charge EVs when demand is low. In local applications, this balancing can help with congestion issues on the distribution grid (Wang and Wang, 2013).

- *Local self-consumption* of renewable energy can be increased with the V2G technology, by charging EVs when the production of renewables is abundant, and discharging when the production is low (Bartolini et al., 2020).
- *Grid balancing* is the concept of maintaining the electrical grid frequency on a stable predefined level by matching the energy demand and supply at every moment of time. Currently, in order to maintain the balance inside the grid, the grid operators employ the services, of so-called, Balancing Service Providers (BSPs). The BSPs are mainly large centralized entities injecting or consuming energy and power and energy in the grid in case of necessity. The V2G-enabled bidirectional charging functionality allows the entities owning and operating V2G charging networks to become decentralized BSPs, consuming and providing balancing energy and power (Li et al., 2021a). On a macro scale, the mass adoption of V2G technology can help to deal with the imbalances created by renewables and thereby allow their share to increase in the energy mix.

Other benefits such as *reactive power compensation* (voltage control) by the charger converter technology have been demonstrated in research (Lenka et al., 2021; Tan et al., 2019), but show no immediate widespread application at the moment.

3. Methodology

Despite the advantages and opportunities enabled by V2G and related technologies, its practical implementation and mass adoption has been very slow due to a number of technical, socio-economic, and other miscellaneous barriers arising in different parts of the value chain. The current paper aims to shed light on the issue via two consecutive steps:

1. **Identification of the barriers** by means of internal expertise, a review of existing literature, and validated by the means of

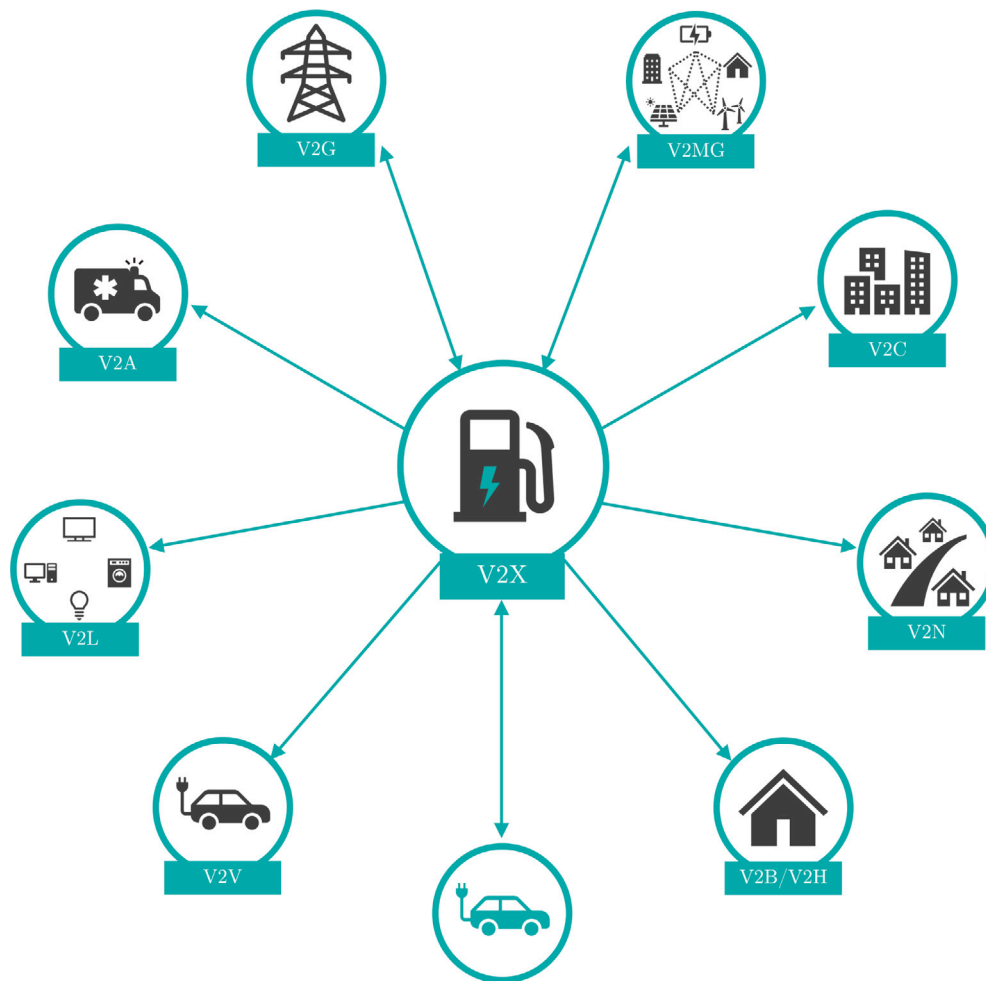


Fig. 2. Different applications of V2G concept.

an external expert survey including 44 experts. The experts participating in the survey represent different types of stakeholders from the EV charging business ecosystem (See Fig. 3), being the potential stakeholders of the V2G-related value propositions (Goncearuc et al., 2022).

As demonstrated in Fig. 3, the stakeholders of the EV charging business ecosystem are divided into three layers. The Core Business part, including direct suppliers and distribution channels of the ecosystem's value proposition, consists of the Charge Point Operators (CPOs), Mobility Service Providers (MSPs), and EV supply equipment (EVSE) manufacturers. The Extended Enterprise part, including the direct customers, suppliers of direct suppliers, and standards bodies, consists of EV users, energy suppliers, TSOs, DSOs, EV manufacturers, EV charging location holders, and the EV charging standards bodies. The upper layer of the business ecosystem has a relatively indirect relation to the ecosystem's value proposition, consisting of various policymakers, regulators, and other indirect stakeholders.

The transformation of the EV charging business ecosystem, triggered by the integration of V2G technology would have an influence on all the participating stakeholders, as a business ecosystem is a dynamic system of interrelated entities (Zhang and Fan, 2010; Scaringella and Radziwon, 2018). Therefore, the expert opinion of the representatives of all these stakeholder groups is highly important for building a realistic overview of the barriers hampering the adoption of the V2G technology.

2. **Analysis of the identified barriers**, investigating the level of risk of non-adoption of V2G technology related to each barrier

and the interconnections between them. As mentioned before, the business ecosystem is a dynamic system of interrelated entities, while the barriers to the adoption of V2G technology are associated with the entities participating in this ecosystem. Thus, these barriers cannot be regarded from an isolated point of view and the understanding of interdependencies between them is extremely important, as the solution of one barrier can facilitate the solution of the others. Therefore, this analysis applies a combination of two popular methods from the literature: (a) the risk matrix (see Section 3.1), assessing the severity and the probability of non-solution of potential barriers, and (b) the Cross-Impact Matrix Multiplication Applied to Classification (MICMAC) (see Section 3.2) (Duperrin and Godet, 1973), analyzing the interrelations between the barriers. Moreover, a new visual representation and MICMAC-based clustering combine the aforementioned aspects of the analysis and facilitate the interpretation of the obtained results (see Section 4.2).

3.1. Risk matrix

In order to assess the risk level of the potential barriers, the current research applies a widely spread risk assessment tool — the risk matrix. The risk matrix consists of two axes: one to account for the likelihood of occurrence of an event and the other for the severity of consequences in case of occurrence. The combination of these two values gives a risk score of an event, ranging from very low to extremely high. The method is very useful for distinguishing the most urgent from the least urgent risks (Garvey P.R., 1998; Bao et al., 2021).

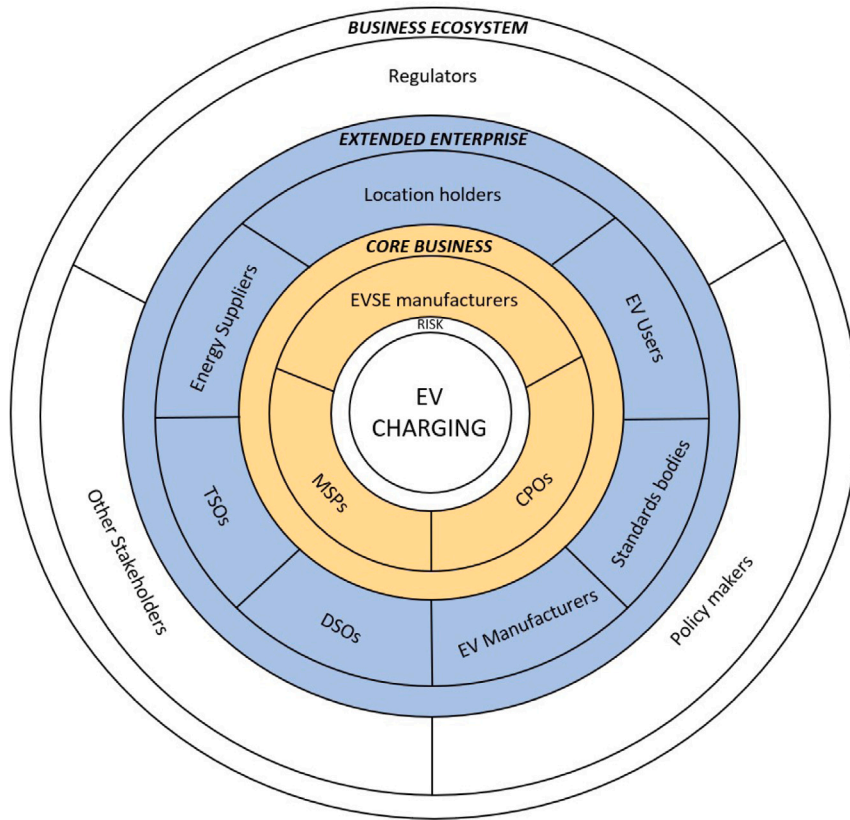


Fig. 3. EV charging business ecosystem (Goncearuc et al., 2022).

In application to the current research, the severity can be interpreted as the level of significance of the barrier today, while the probability, is the presumed chance that the barrier would not be solved in the near future. It should be pointed out once more, that the current research uses the reverse logic of the problem statement, evaluating the likelihood that the barrier will not be solved within a relatively short term, along with the severity of this outcome. The risk score, is, thus, a derived variable, indicating, in this case, the most severe barriers, that are difficult (or unlikely) to be solved.

Based on the latter, all the potential V2G barriers are evaluated by the experts' survey composed of different stakeholders from the EV charging business ecosystem (Goncearuc et al., 2022), demonstrated in Fig. 3. The likelihood and severity scores of the barriers are evaluated on the scales from 0 (not likely, not severe, respectively) to 5 (extremely likely, extremely severe, respectively). The risk score is retrieved by the multiplication of likelihood and severity scores. This results in derived 0 to 25 risk scores, aggregating the impact of each potential barrier on all the potential stakeholders (see Fig. 4).

3.2. Matrice d'impacts croisés multiplication appliqués à un classement (cross-impact matrix multiplication applied to classification) (MICMAC)

In order to assess the potential influence of one barrier on another, the driving power and strength of interrelations between the barriers, the current research applies the Cross-Impact Matrix Multiplication Applied to Classification (MICMAC) method, developed by Duperrin and Godet (Duperrin and Godet, 1973). The MICMAC method includes three consecutive steps (Hu et al., 2009):

1. **Definition of the relevant factors.** This step of MICMAC analysis is equivalent to Step 1 of the methodology applied for the whole current research — the **Identification of the barriers** to the V2G adoption.

2. Determination of interrelations between the defined factors.

The current study uses the Matrix of Direct Influence (MDI) as a tool for the description of interrelations between the V2G barriers. The MDI can be described as an $n \times n$ matrix, where n is the number of defined barriers:

$$MDI = \begin{bmatrix} 0 & x_{12} & \dots & x_{1z} & \dots & x_{1n} \\ x_{21} & 0 & \dots & x_{2z} & \dots & x_{2n} \\ \dots & \dots & \ddots & \dots & \dots & \dots \\ x_{y1} & x_{y2} & \dots & 0 & \dots & x_{yn} \\ \dots & \dots & \dots & \dots & \ddots & \dots \\ x_{n1} & \dots & x_{nz} & \dots & x_{n\ n-1} & 0 \end{bmatrix}$$

for $x_{yz} \in \{0, 1, 2, 3\} \mid x_{yz} = 0 \forall y = z, y \in \{1, \dots, n\} \wedge z \in \{1, \dots, n\}$

Every x_{yz} is the value within MDI, indicating the strength of the direct influence of barrier number y on barrier number z on a scale from 0 (no influence) to 3 (high influence). The values for filling the MDI in the current study are retrieved by means of internal expertise and the review of the existing relevant literature (Duperrin and Godet, 1973) and are provided in Appendix. The assessment of the strength of influence of every barrier on the rest of the defined barriers generates an $n \times n$ matrix. Every row of this matrix represents all the influence scores of one barrier to V2G adoption on the others and every column represents all the dependence scores of one barrier on the others. The MDI diagonal ($y = z$) indicates the influence of the barrier on itself and is set to a 0 value.

3. **Identification of key factors.** The final step of the MICMAC analysis involves building the Influence - Dependence map, based on two types of MDI results. The first type is the sum of the values of every MDI row, indicating the level of influence (driving power) of the barrier corresponding to this row on all the other barriers. The second type of MDI result is the

sum of the values of every MDI column, indicating the level of dependence of the barrier on all the other barriers.

Putting the defined dependence and influence of the barriers in a two-dimensional environment allows to build an Influence - Dependence map and group the V2G barriers into four clusters:

I. Autonomous. Barriers that are relatively unrelated to all the other barriers. The autonomous barriers have a low level of influence, but are also weakly dependent on the changes of other barriers. Therefore, they should be tackled in an individual manner, without relating the potential results to the changes of other barriers.

II. Dependent. Barriers that have a relatively low influence, but are strongly dependent on the changes of other barriers. In general, the dependent barriers can be considered as the resulting ones, originating from the other defined barriers. Therefore, their solution is possible only in case the original barriers strongly influencing the dependent ones are solved.

III. Linkage. Barriers that have a high influence but are also strongly dependent on the changes of other barriers. Generally considered as the most unstable barriers, representing interesting opportunities as the intermediate link into the potential impact chain.

IV. Driver. Barriers that have a high driving power and strongly influence other barriers but are, at the same time, weakly dependent on the changes of other barriers. The driver barriers are the most important barriers from the inter-relational point of view as their independence cuts the opportunity to be solved indirectly through other barriers, while the high level of impact is decisive for the solution of the barriers from the previous two clusters.

Finally, as a combination of the above-described methods, the evaluation parameters of the risk matrix are plotted as a third dimension (combination of color and size) in the MICMAC Influence-Dependence map (see Section 4.2).

4. Results

4.1. Identification: current barriers to V2G adoption

This section defines and describes the 23 potential barriers to the adoption of the V2G technology.

Barrier №1: Low EV penetration rate

The EV penetration rate is about to experience exponential growth (IEA, 2024; Elia group, 2020), but has not yet reached the mass adoption phase, which would allow for economies of scale and a decrease in prices. Being a complementary good, V2G technology, consequently, is not able to spread either. Moreover, the EVs would require an additional update in order to become V2G capable (See *Barrier №: 21*).

Barrier №2: Complex technology for V2G chargers

The existing direct current (DC) V2G chargers on the market are significantly more complex than unidirectional chargers. It includes many additional power electronics components, which make the charger heavier, less environmentally friendly and more expensive (Yu et al., 2022). An alternative might be the technology allowing for alternating current (AC) bidirectional charging, requiring, however, costly software and hardware upgrades on the EV side.

Barrier №3: Complex optimizations for energy management system

Bidirectional charging can quickly become a complex Energy Management System (EMS) optimization problem when considering many different parameters such as maximum and minimum powers, state-of-charge (SOC), efficiency losses, congestion, load of buildings, EV user preferences, EV models, etc. Moreover, the adoption of V2G technology

can potentially help to deal with the volatility of renewable energy generation and increase its share in the energy mix. However, the disordered charging and discharging of a large number of EVs would not only fail to address the renewables' volatility, but also aggravate the fluctuations of loads (Li et al., 2021b). Therefore, the coordination of charging and discharging activities becomes another EMS optimization challenge. The complexity of this challenge also increases with the number of assets to coordinate. The more EVs and chargers to control and the longer these EVs remain plugged in, the bigger the computing complexity is (Yi et al., 2020). This complexity thus already manifests itself on a local scale but becomes much more challenging for entities managing or interacting with larger networks of chargers (CPOs and grid operators) (Noel et al., 2019c; Salvatti et al., 2020).

Barrier №4: Competing technological solutions

Recent studies (Van Krieking et al., 2021; Mukherjee et al., 2021) show that for specific services (see Section 2), uni-directional smart charging can have similar results to bidirectional charging, limiting the added value of V2G. Even though the V2G technology is a tool with broader functionality (allowing EVs to consume and supply energy and power back to the grid), unidirectional smart charging is a less efficient but much cheaper option (See *Barrier №15*).

Moreover, the grid operators, being the main potential beneficiaries of the V2G-enabled grid balancing services, are reluctant to the adoption of the V2G technology as the traditional V2G substitutes for grid reinforcement are feasible and well-known (e.g. advanced system control devices, grid forming converters, line and voltage upgrades, etc.) (WindEurope, 2020).

Barrier №5: New and complex communication standard between electric vehicles and chargers

A new standard (ISO15118) for the communication between EV and charger has been defined and is being implemented by EV and EVSE manufacturers (Anon, 2019a). However, as with any new standard, it requires considerable resources to develop this from the ground up. The ISO15118 is a standard complementing the current European agreed dominant standard of the CCS-plug, in response to the Japanese CHAdeMO standard. Though CCS and ISO15118 are agreed to be the standard for Europe, it has not yet become mandatory, and the existence of a multi-standard for DC charging puts pressure on EVSE manufacturers and slows down the process due to dispersed focus. Especially, considering CHAdeMO has an early mover's advantage, already having the developed V2G chargers and V2G-capable EVs (ex. Nissan Leaf, Mitsubishi Outlander) on the market (Wang et al., 2014).

Barrier №6: Communication standard to control chargers

The bidirectional chargers require a new line of communication in order to be controllable. Currently, the most common communication protocol is called OCPP (Open Charge Point Protocol) 1.6 (Anon, 2023a). However, its new version (OCPP2.0.1) is already under construction, aiming to allow monitoring of advanced parameters of the EV and to send commands to the chargers (Anon, 2023a). Similar to *Barrier №5*, such a new version requires considerable resources to be implemented by the control entity, be it TSOs, DSOs, CPOs, aggregators, LES (Local Energy System), or others (De Cauwer et al., 2021). Furthermore, it is not yet certain whether this protocol would be dominant and commonly accepted, as other rival protocols are currently under development (IEC 63110, OpenADR or EN 50491-12-2).

Barrier №7: Cyber Security

According to Hoang et al. (2017), the provision of V2G services has the following of the cyber security requirements:

- **Confidentiality** of the data stored by the V2G services organizers (e.g. operational settings of the system, control commands, clock settings, applied energy prices, charging/discharging data) (Han and Xiao, 2016). The lack of confidentiality mechanisms makes the service organizer vulnerable to competition and reduces the willingness to participate in V2G-related services.

- *Authentication* mechanisms for the EVs participating in V2G, customized and specified for every EV type and model.
- *Integrity*, ensuring the security of sensitive data from the potential modifications from the third party.
- *Availability*, facilitating the secure data exchange and communication between the actors of the V2G value chain.

The non-fulfillment of these cyber security requirements and the additional cyber–physical elements that the V2G technology brings to the electricity grid increase its vulnerability and make it prone to various attack scenarios (e.g. malicious command injections, corrupted data injection, communication delays) (Saxena et al., 2017). The interdependence that the V2G brings between the energy stored in the EVs and the electrical grid can lead to grid disturbances, in case the former becomes the target of a massive malicious command injection attack. As a result, grid operators together with CPOs and MSPs are the entities mostly impacted by the constraints that cyber security poses. The first steps towards ensuring the cyber security of V2G applications were already made by the introduction of ISO 15118 standard. However, as it is described in barrier №5, its mass implementation and continuous update according to the appearing cyber security threads, still require significant efforts from the stakeholders.

Barrier №8: Privacy

In V2G networks the EV user can assume two roles and act both as a customer and as a service provider. In both cases, information, such as customer identification, location and payment/tariff data, access control policies (Han and Xiao, 2016), as well as other vehicle-related preferences (e.g. SoC of the battery and charging/discharging pattern), constitute a large set of privacy-sensitive data. All these render the EV user a vulnerable stakeholder to privacy-related issues.

More specifically, three unique characteristics of the V2G technology, the dynamic participation of EVs in the V2G scheme (i.e. connect and disconnect to and from grids without influencing ongoing communications), the EV mobility (i.e. ability to connect to various electric grids by changing location) and the role-based vehicle and payment system (Saxena et al., 2017), pose big challenges to the privacy-preserving charging and discharging data publication (data made available online to users for their own use and statistical reference) (Han and Xiao, 2016).

Barrier №9: Range anxiety

The shorter range of EVs compared to ICE vehicles is often seen as a barrier by potential EV buyers (Rezvani et al., 2015). Since EVs are already perceived as having a limited range, the potential participation in V2G-related services consequently increases the EV users' concerns about the minimum SOC (especially in the case of spontaneous trips) (Geske and Schumann, 2018; Huang et al., 2021; van Heuveln et al., 2021). Considering the fact that EV users' perception and satisfaction are the dominant factors influencing the sales of EV manufacturers, V2G-related range anxiety might trigger discontent and can explain the relatively slow pace of European EV manufacturers to adopt V2G capabilities in their vehicles. A similar reasoning applies to CPOs and MSPs, directly responsible for the successful delivery of the charging energy and providing the required range for the driver (Esmaili et al., 2018).

Barrier №10: Lack of understanding

The V2G technology is relatively new and may be perceived as a complicated system by the EV user. Therefore, it becomes difficult to understand the process and value of this technology, which hinders the EV users' willingness to use the system (Noel et al., 2019a). Optimistically, the EV users may simply see V2G as an "electric grid affair" (Kester et al., 2019), while in more extreme cases, they might not even be aware of the existence of the technology (Geske and Schumann, 2018). Therefore, it becomes important for the adoption of V2G to create a clear and transparent business model for V2G-related services (Noel et al., 2019a) and increase awareness about the

technology. This brings a challenge to stakeholders reaping from V2G applications to inform EV users of its benefits for society and provide clear financial incentives for participation.

Barrier №11: Control over the charging process

Electric vehicle users have different preferences concerning the level of control they expect from a V2G system. Some prefer full automation whereas others require (partial) control (Kester et al., 2019; van Heuveln et al., 2021). Additionally, providing battery access to a third party might not always be appreciated by the end-user (Noel et al., 2019a). Similarly to range anxiety, this impacts both the EV users' willingness to participate in V2G-related services and overall acceptance of the vehicle. Certainly, the customers' perceptions strongly influence the decisions of EV and EVSE manufacturers on whether or not to engage themselves in the production of V2G-capable EVs and V2G EVSE respectively.

Barrier №12: Battery degradation

So far, the existing literature has not come to a consensus about the V2G impact on the additional degradation of EV battery. While various researches found no additional wear-off (Baure and Dubarry, 2020; Li et al., 2018; Thingvad and Marinelli, 2019), other studies report additional battery degradation related to V2G participation (Thingvad et al., 2021; Uddin et al., 2018; Wang et al., 2016; Bishop et al., 2013; Dubarry et al., 2017). The important observation is that the additional degradation is dependent on the battery type and the type of V2G service provided (Guo et al., 2019).

It is also important to differentiate the acceptance of additional battery degradation between EV owners and EV lessees. Therefore, the solution could be the creation of an adequate and flexible compensation mechanism, matching the expectations of various groups of stakeholders (van Heuveln et al., 2021). However, the problem is not yet widely studied and requires more practical validation.

Barrier №13: Uncertain V2G profits for EV users

The EV users participating in V2G-related services expect to be compensated both for the provided service and presumed battery degradation (Noel et al., 2019a). The current small-scale, mostly project-initiated, V2G EVSE landscape, where EV users can engage in a variety of energy services, does not offer a clear compensation mechanism (Huber et al., 2021). In addition, the EV users' expectations about the amount and form of compensation remain diverse as well (van Heuveln et al., 2021). As a consequence, the variety of different expectations in combination with a lack of transparency creates an area of uncertainty for EV users and decreases their willingness to adopt the V2G technology.

Barrier №14: Long value chain

The introduction of the V2G technology in the EV charging business ecosystem (see Fig. 3) implies the simultaneous collaboration of multiple entities forming a new value chain Goncearuc et al. (2022). According to Noel et al. (2019b), the operation of the V2G-related service value chain requires four types of interactions between its participants: financial flow, bidirectional power flow, communication and regulations. The longer the value chain, the more complex these interactions become.

The uncertainty about the potential profits, that have to be shared between all the entities in the value chain, leads to insufficient financial stimulus for participation (Huber et al., 2021). Moreover, the challenges related to interoperability between different participants of the value chain, common communication and power transfer standards are aggravated by the rival needs of these participants (e.g. "vendor lock-in" standards that commit customers to one service provider) (Anon, 2019b).

Barrier №15: Cost of V2G chargers

The potential of V2G capable chargers in smart charging/energy services is significantly higher, compared to unidirectional chargers.

However, the current dominant technology for V2G chargers is based on DC-chargers, which are significantly more expensive than the AC chargers of equivalent power level, because they require additional power electronic components (see *Barrier N2*). The average market price of a DC V2G charger multiplies the price of an unidirectional AC charger of an equivalent power level by a factor of 5–10. The limitations in reducing the production cost make it very difficult for CPOs and location holders to apply V2G in a cost-efficient and attractive, to the EV user, way (*Steward, 2017*).

The AC V2G technology already exists and can make V2G cheaper by using power electronics already present in the vehicle. However, that requires adoption from the vehicle side and has some remaining technological challenges (*Steward, 2017; Anon, 2023b*).

Barrier N16: Difficult energy trading and balancing markets entry

The V2G technology, shows a relatively high potential for TSO/DSO grid balancing, along with energy trading on the imbalance markets (*Elia Transmission Belgium, 2021b,a*). However, in the European Union (EU), traditionally, these services were provided only by large entities with a high power capacity and a limited number of energy delivery points. This explains the rather centralized grid-balancing market structure with relatively high entry barriers (*Anon, 2020, 2017; Elia, 2020a, 2019, 2020b*):

- Minimum capacity bid: the minimal contractual power capacity for grid balancing services is 1 MW.
- Minimum energy bid: the minimal energy capacity for trading on energy imbalance markets in Belgium is 1 MWh.
- Expensive metering equipment: participation in grid balancing services and energy imbalance markets requires precise certified metering equipment. This metering equipment should be installed on every energy delivery point, in this case V2G chargers, further increasing the total costs of V2G chargers, which are already quite high (see *Barrier N15*).

These restrictions, and a lack of a clear attractive financial stimulus, make the potential organizers of the V2G-related services quite reluctant to engage themselves in this business (*Mojumder et al., 2022*).

Barrier N17: Interaction with parking for charging

The use of V2G and smart charging for energy balancing services implies long EV plug-in and parking periods, increasing the occupation rate of chargers and parking spots. Considering the business model design of the major part of the core participants of EV charging business ecosystem (*Goncearuc et al., 2022*), where the revenue streams are mainly based on the amount of the transferred energy, long plug-in sessions after the end of the charging process reduce the charging revenues of CPOs/MSPs and location holders. Therefore, the revenues from the provision of energy services should at least cover these missing charging revenues. Additionally, this can have a negative impact on the willingness of the customer to participate by feeling constrained (*Huang et al., 2021*). All this makes the integration of V2G-enabled services less attractive in application to public EV charging infrastructure.

Barrier N18: Absence of a business case for organizers

The potential V2G-related services organizers, being entities in the middle of the value chain (CPOs, MSPs, and location holders) (*De Cauwer et al., 2021; Goncearuc et al., 2022*), are not able to efficiently assess the potential benefits and risks of incorporating the V2G-related services into their current business models due to the lack of benchmarking opportunities. Moreover, a number of challenges arises from the process of introducing V2G as a new technology into an existing business model of a company (*Goncearuc et al., 2022*):

- Adjustment of *key resources*, requiring significant investments (e.g. V2G EVSE infrastructure, specialized software, specialized HR, etc.);

- Introduction of new *value propositions* related to grid balancing and energy trading services, which come with relatively high entry barriers (see *Barrier N16*);
- Attraction of new *customer segments*, which are not necessarily willing to purchase the newly offered *value propositions* due to the existing substitutes (see *Barrier N4*). For instance, the TSOs and DSOs, which could become the new customers, currently show more loyalty to the traditional grid reinforcement techniques, while the EV users are uncertain about the profitability of their participation in the V2G-related services as partners (see *Barrier N13*).
- Unclear taxation rules, negatively affecting the expectations about the potential new *revenue streams* (see *Barrier N19*).

As the profitability of CPOs, MSPs and location holders from V2G services remain questionable, the absence of a business case will propagate into other barriers thus creating a chain of cascading side effects.

Barrier N19: Taxation

The EV charging business ecosystem does not have a clear understanding of the boundaries and process of V2G tariff formation, due to the lack of clear V2G tariffication and taxation rules. Fiscal-wise, V2G is currently seen as an ordinary energy flow, but not as a flexibility service (*PwC, 2019*). Therefore, the energy taxes remain equal for charging and re-charging the previously discharged energy, leading to unintended double taxation.

The rigidity of energy taxation schemes and lack of V2G-specific rules hampers the opportunity for the CPOs and MSPs to engage EV users in flexibility services by offering them favorable tariffs. At the same time, participation in V2G-related services would change the EV users' status from energy consumers to prosumers, which should be fiscally treated as suppliers of balancing power and energy to the grid (*Interreg, 2020; Kester et al., 2018; PwC, 2019*).

Barrier N20: Communication between charging services actors

One of the main objectives of coordinated charging is the reduction of charging-related expenses, which are strongly influenced by the choice of charging method and technology (uncoordinated, smart charging, or V2G). Different pricing mechanisms related to each of the charging options should be communicated to EV users, requiring the modification of existing charging services communication protocols (OCPI, IEC63119, OCHP, OICP, etc.) (*Hussain et al., 2018*).

Barrier N21: Lack of V2G-capable EVs

The majority of EVs require software and communication developments in order to make them compatible with V2G technology. Despite the fact that a number of EV manufacturers has already launched the production of V2G-capable EVs (e.g. Nissan, Mitsubishi, Ford, Volkswagen, Hyundai, KIA, MG), currently, most of the EVs on the market are not V2G-capable. This shortage and consequential lack of large-scale testing hamper further research and developments, potentially able to reduce the costs and/or generally improve the V2G technology. At the same time, there is no clear incentive for EV manufacturers to allow for the discharging of their EVs, as it currently brings no added value to their product and might affect customers' loyalty due to the potential third-party battery access and battery degradation (see *Barriers N15 and 16*).

Barrier N22: Failure to secure a critical mass of early adopters of V2G

The V2G technology is a networking good strongly dependent on the critical mass of the technology users. For this type of goods, the technology adoption, after reaching the critical mass of early adopters and becoming familiar to a broader public, could be represented by an exponential curve until it reaches the market potential (*Grajek and Kretschmer, 2008*). In practice, it means that the more users adopt the V2G technology, the faster a larger number of new users are attracted. However, currently, the V2G technology has not yet reached this critical mass of early adopters triggering further exponential growth.

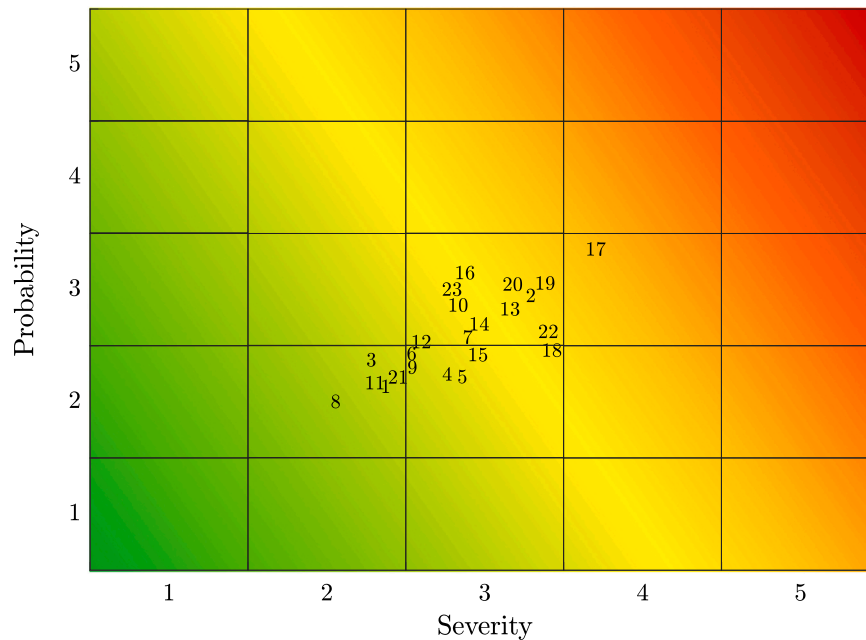


Fig. 4. Risk matrix for the barriers to V2G adoption.

Barrier №23: Lack of large-scale demonstrations

The difficulty in organizing and managing large-scale demonstrations could be a barrier to the integration of smart charging technologies according to the European Network of Transmission System Operators for Electricity (ENTSO-E) (Anon, 2021b). This could be also applicable to V2G technology with the main challenges being the following. First, the consistent participation of a large number of users needs to be ensured for a significant period of time in a sufficiently large space which can also be difficult to secure. Moreover, the large capital investment that such demos require can discourage market parties from participating and investing. Finally, the coordination of all the involved stakeholders can prove to be a real hurdle.

4.2. Analysis

After the identification of the barriers hampering the adoption of V2G technology, it becomes possible to analyze them from different angles and points of view. The first result of the analysis is the Risk Matrix, presented in Fig. 4. The Risk Matrix is retrieved by means of the survey of the opinions of 44 experts from different stakeholder groups of EV charging business ecosystem (see Fig. 3). The participating experts evaluated the probability (of not being solved) and the severity of the identified barriers on a scale from 0 to 5. The multiplication of the defined probability and severity scores resulted in a derived risk score of 0 to 25 for each of the barriers. The resulting risk scores with their level of probability (on the y-axis) and severity (on the x-axis) are presented in Fig. 4. Moreover, Fig. 4 visualizes the risk level of every barrier through the color codes, where green indicates the low level of risk, yellow — medium, and red — high.

As it is noticeable from Fig. 4, the highest concentration of barriers is in the yellow area of the Risk Matrix, within the medium risk score range. This is a promising result, showing that on the one side, the experts acknowledge the significance of the barriers, not finding, however, any extremely severe issues (severity score ranges from 2.1 to 3.7 out of 5). On the other side, the experts did not point out any of the barriers as unsolvable (probability score ranges from 2 to 3.2 out of 5), which results in a moderate level of risk for the whole population of the barriers.

The next important result of the analysis is demonstrated in Fig. 5, which combines the information contained in the Risk matrix (see Fig. 4) with the results from the MICMAC analysis (see Section 3.2). The results of the MICMAC analysis are retrieved by the means of expertise about the influence and dependence of every identified barrier on the others. The influence/dependence were evaluated on a 0 (no influence/dependence) to 3 (high influence/dependence) scale. Thereafter, the aggregated influence and dependence scores of every barrier were put in the MDI map (Fig. 5), demonstrating their degree of dependence on the x-axis, and degree of influence on the y-axis. Furthermore, the third dimension has been added in the shape of the size and color of the data point of each barrier, where size and color code indicate the overall risk score, as a result of the Risk matrix analysis.

The barriers that tend to be low on the dependence scale, indicating that these are to be tackled separately/individually and will not disappear by solving other barriers, are mainly the barriers linked to technological challenges of V2G (e.g. №2: *Complex technology for DC EVSE*, №3: *Complex optimization for energy management system*, №12: *Battery aging*, etc.) and V2G-related business issues (e.g. №15: *cost of V2G EVSE*, №16: *Difficult energy and balancing markets entry*, №19: *Taxation*, №17: *Interaction with parking for charging*, №14: *Long value chain*, etc.). The only business-related exception showing a high level of dependence is the barrier that could be considered as resulting from the other components of the V2G-related business model - №18: *Absence of the business case for the organizers*.

From the severity, probability, and, thus, the derived risk points of view, the most prominent barriers are also the ones related to the V2G business modeling issues, mainly representing the upper boundary of the medium risk level. However, it is important to mention that the technological barrier №2: *Complex technology for DC EVSE* shows the upper-medium risk level as well.

On the opposite, the highly dependent barriers, are mainly the ones linked to the regulative standards (e.g. №6: *Communication standard to control chargers*, №5: *New and complex communication standard between EVs and EVSE*) and EV users. The highest risk scores among the dependent barriers group are found on the EV users' side (e.g. №22: *Failure to secure a critical mass of early adopters*, №13: *Uncertain V2G profits for EV users*, №10: *Lack of understanding*, etc.). According to Appendix, these EV users-related V2G barriers are, among others, strongly influenced by

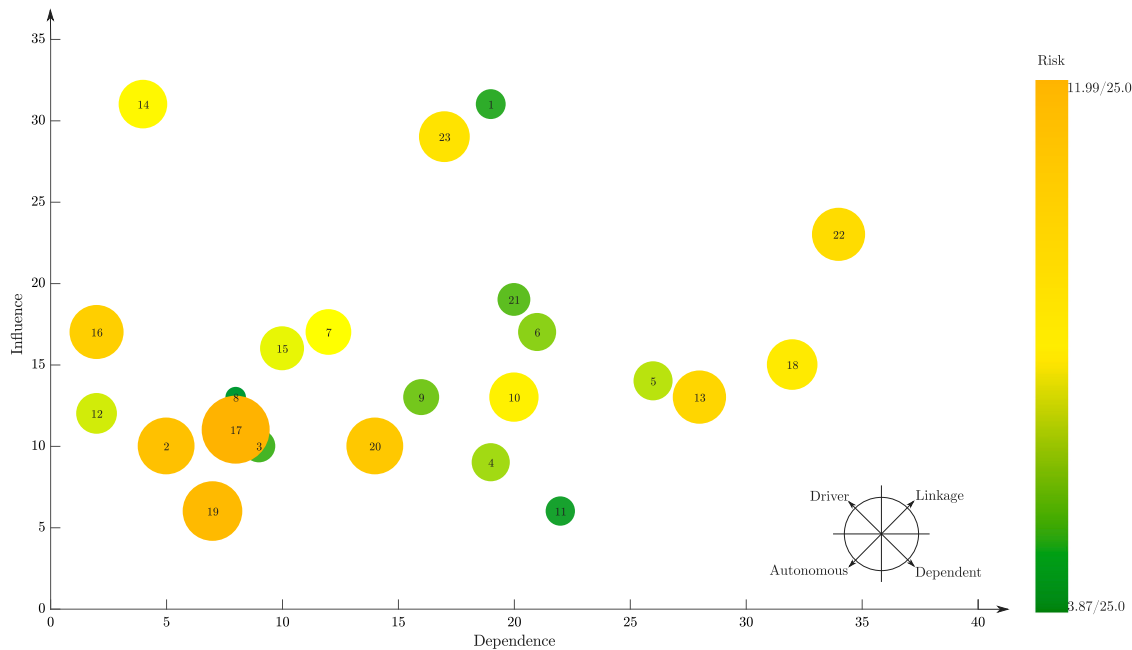


Fig. 5. MDI map with risk score indication. The color and size scales are one-to-one with the attributed scores to be in accordance with Fig. 4. The score 11.99 indicates the maximum calculated risk score of a barrier. The maximum theoretical score is 25.

barriers №18: *Absence of the business case for the organizers*, №23: *Lack of large-scale demonstrations* and №21: *Lack of V2G-capable EVs*. Therefore, the solution of these influential barriers can significantly facilitate the above-mentioned EV users-related hurdles to V2G adoption.

At the same time, some of these EV users'-related barriers combine the high level of dependence with the high level of influence, namely №21: *Lack of V2G-capable EVs* and №22: *Failure to secure a critical mass of early adopters* of V2G. These barriers relate to the recent appearance of the V2G market and could be potentially solved by the growth in its maturity level, significantly accelerating further growth. Even though the existence of these barriers looks like a chicken-and-egg deadlock at first sight, their high level of dependence shows that they can be (at least partially) solved by solving other barriers. This is also fueled by the experts' belief that with increasing battery capacity, availability of V2G infrastructure, and initial V2G experience of a significant quantity of EV drivers, those barriers would diminish (Kumar and Alok, 2020).

Finally, the results of this analysis show three important barriers that show the highest level of influence. The first barrier in this list is barrier №14: *Long value chain*, having a high influence and a very low dependence on the other barriers. Although the risk score of barrier №14 is medium, all combined, the analysis shows that the barrier is important. The solution to this barrier does not necessarily imply the reduction of the value chain, but its smooth operation. This requires significant efforts and simultaneous collaboration of all the stakeholders, defining clear and transparent V2G revenue models, and establishing interoperable communication and charging/discharging standards. It is important to mention, that the long value chain challenges have a high influence on barriers №5, 7, 13, 20, 23, along with medium and low influence on other barriers (see Appendix). Therefore, the solution of barrier №14 would facilitate the solution of numerous other barriers.

The second is the barrier №23: *Lack of large-scale demonstrations*. This barrier has a high level of influence and a medium level of dependence, combined with a medium risk level (although slightly higher than barrier №14). With this particular set of characteristics, the large-scale V2G demonstrations are therefore identified in this ecosystem as a good enabler to overcome other barriers as well. Moreover, the first examples of large-scale V2G demonstration projects do already exist (Amsterdam

University of Applied Sciences, 2020), while the European and National research and innovation funding schemes (Anon, 2022) give the prospects of more future projects and developments in this direction.

The third influential barrier is №1: *Low EV penetration rate*, having a very high level of influence and a medium level of dependence, combined with one of the lowest risk scores in this ecosystem. However, the EV market promotion receives a lot of attention and dedication at the moment. Considering the latest European targets and ICE vehicle future bans (European Council, 2024a), the EV penetration rate, along with related infrastructure and other complementary products, is anticipated to disappear as a barrier to V2G adoption in the near future, which explains the lower perceived risk level.

As mentioned in the last step of MICMAC analysis (see Section 3.2) the barriers from the MDI map demonstrated in Fig. 5 can be clustered by their level of dependence and influence. The median influence and dependence scores of the barriers define the borders, splitting the MDI map into four clusters. The clusters with an above-median influence score have a higher priority, as their solution may facilitate the solution of the dependent barriers and boost the whole process. Consequently, the clusters of the barriers with the above-median dependence score have a lower priority, as they could be (partially) solved by solving the more influential barriers. The four clusters of the barriers, along with the respective risk scores are visualized in Fig. 6.

It is important to point out that the structure shown in Fig. 6 is not an absolute priority list and the simultaneous collaboration of stakeholders in different directions is essential for the V2G adoption. The structure provided by Fig. 6 aims to create an understanding of how to facilitate and optimize the solution process.

Cluster IV: Driver is on the top of Fig. 6, indicating the barriers with low dependence and high influence on the other barriers. In other words, solving the IV: *Driver* barriers, which are relatively independent of the other, would also help to (partially) solve many other barriers from the list. Inside the IV: *Driver* cluster the barriers have moderate risk scores, neither hitting the upper nor the lower boundaries of the medium risk level. It is also noticeable that these barriers are mainly business-related issues. Considering their high importance and influence, combined with the low level of dependence, there is a need

MDI cluster	Nº	Barrier name	Risk score
IV: Driver	16	Difficult energy and balancing markets entry	8.70
	14	Long value chain	7.62
	7	Cyber Security	7.10
	15	Cost of V2G EVSE	6.79
↓			
III: Linkage	22	Failure to secure a critical mass of early adopters	8.52
	23	Lack of large-scale demonstrations	8.04
	6	Communication standard to control chargers	5.86
	21	Lack of V2G-capable electric vehicles	5.14
	1	Low EV penetration rate	4.78
↓			
II: Dependent	13	Uncertain V2G profits for EV users	8.55
	18	Absence of business case for organizers	8.01
	10	Lack of understanding	7.76
	5	New and complex communication standard between EV and EVSE	6.01
	4	Technological solution of competitors	5.87
	11	Control over the charging process	4.72
↔			
I: Autonomous	17	Interaction with parking for charging	11.99
	19	Taxation	9.93
	2	Complex technology for DC chargers	9.30
	20	Communication between charging services actors	9.29
	12	Battery aging	6.27
	9	Range anxiety	5.56
	3	Complex optimization for energy management system	5.14
	8	Privacy	3.87

Fig. 6. MDI Clusters of barriers with risk scores indication.

for external effort for their solution, facilitating the solution of the other barriers.

For instance, the solution of the barrier with the highest risk score in this cluster, №16: *Difficult energy and balancing market entry*, is an enabler for market access and profit generation. At the same time, its level of dependence is the lowest in the whole ecosystem, meaning that even solving all the other barriers would barely help with this one. Therefore it requires the first move from the regulative and legislative side. The successful integration of the V2G technology in the energy and balancing markets requires a change in the design of these markets. This market design can, according to Ismail et al. (2023), be summarized in a so-called, 5D vision, comprising the Decentralization of energy-producing and grid-balancing entities; De-carbonization of energy production and use of renewables; Digitalization of energy-related data, metering, and processes; Deregulation and Democratization of the energy and balancing markets increasing the internal competition. The recent years have shown certain developments in the direction of implementation of this 5D vision in the EU. The increased use of renewables, installation of digital meters, deregulation of energy supply, and other important steps in this direction are already observed (Anon, 2021c). However, the remaining regulatory barriers still cut the opportunity for V2G business ecosystem stakeholders to participate in energy and balancing markets.

Cluster III: *Linkage* is the second on the priority list. Some of the barriers from this cluster might be (partially) solved by solving the IV: *Driver* cluster barriers, as the barriers from III: *Linkage* cluster show a high level of dependence. At the same time, these barriers show a high level of influence as well, meaning that solving these barriers would facilitate the solution of the other barriers.

The risk scores in this cluster vary from the lower to upper boundaries of the medium risk level. The barriers with the lower risk score

in this cluster are mainly the ones with the initiated solution process, indicating the positive perceptions of stakeholders. For instance, overcoming №1: *Low EV penetration rate* is an important enabler and complementary good of V2G technology. This barrier is currently being actively diminished and the EV market shows significant growth (as mentioned before), having therefore a lower perceived risk score. However, being also dependent on a number of barriers within the defined list, the EV penetration rate can show a significant boost by solving the barriers preventing the adoption of V2G technology. The V2G-enabled services can eventually become another important or even decisive benefit of owning an EV.

The higher risk scores within the III: *Linkage* cluster, for barriers №22: *Failure to secure a critical mass of early adopters* and №23: *Lack of large-scale demonstrations*, mainly originate in the risks related to the barriers on which they are dependent. Therefore, solving the barriers from Cluster IV significantly facilitates the organization of large-scale V2G demonstrations (№23), which would allow to reach the critical mass of V2G early adopters (№22).

Cluster II: *Dependent* is composed of barriers with a high level of dependence and low influence. In general, this cluster has the lowest priority as this group of barriers, regardless of its risk scores, is likely to be (at least partially) solved by solving the barriers from clusters IV: *Driver* and III: *Linkage*.

Barriers №13 and №18 are very representative examples of the II: *Dependent* cluster. On the one side, these barriers have relatively high risk scores in comparison to the rest of the ecosystem, but on the other side, these risk scores originate in the other barriers influencing them. As mentioned before, barrier №18: *Absence of the business case for the organizers* is highly dependent on all the components of the V2G-related business model, including barriers №16, 14, 15, 22, 23 and others. A

similar reasoning is valid for barrier №13: *Uncertain V2G profits for EV users*.

Cluster I: *Autonomous* contains the barriers low both on dependence and influence levels. The barriers from this cluster should be solved independently and in parallel to the barriers from the previous three clusters. Considering the fact that this cluster contains a group of barriers with the highest risk scores within the whole ecosystem, it gives them a very high priority. It is important to notice that I: *Autonomous* cluster contains both the lowest and the highest risk scores of the ecosystem — barriers №8 and №17. The lowest risk score of the barrier №8: *Privacy* can be explained by the existence of numerous benchmarking opportunities and significant expert knowledge on privacy protection mechanisms.

The highest risk score of the barrier №17: *Interaction with parking for charging* is a more intriguing result. According to the definition of barrier №17 (see Section 4.1), the V2G-related value propositions increase the EV plug-in periods, decreasing turnover and diminishing the total amount of energy per charger sold to EV users. Thus, from a generalized point of view, the integration of V2G technology into the business model of charging services organizers creates a cannibalizing value proposition, with uncertain chances of success. Therefore, it becomes clear why the stakeholders mainly representing the existing EV charging business ecosystem find it very risky to transform the existing Parking-for-charging business model by means of the inclusion of V2G-enabled services.

However, according to Appendix, barrier №17, while being in I: *Autonomous* cluster for its aggregated influence and dependence scores, is still highly dependent on one particular barrier, namely barrier №18: *Absence of the business case for the organizers*. Therefore, barrier №17 could be solved by showing that the V2G-enabled services can be integrated into the business models of the EV charging business ecosystem in a profitable and non-cannibalizing manner. An example of this complementary integration is the niche targeting of the charging spots with a high occupation rate and a low turnover. Those are the charging locations where EV users naturally spend a significant amount of time per charging session, and the plug-in periods are longer than necessary to fully charge their EV batteries (e.g. business centers, manufacturing sites, hospitals, large residential buildings, etc.). This approach would allow to maintain the existing charging revenues from these locations while creating an additional revenue stream related to the provision of V2G-related value propositions (Goncearuc et al., 2023).

5. Conclusions

From the performed research, it can be concluded that the adoption of V2G technology is a complex issue that still has a considerable number of significant barriers to overcome, requiring the collaboration of different stakeholders. The current paper has identified 23 barriers potentially hampering the adoption of V2G technology, including technical, business, and EV users-related challenges. Furthermore, the analysis of the identified barriers has defined the risk for non-adoption of V2G technology associated with every barrier, along with the influence and dependence of every barrier on the others.

From the performed analysis of the identified barriers, it becomes clear that the barriers related to EV users show a medium level of risk, indicating them as a relatively important point of attention for the adoption of V2G technology. However, their medium level of risk is also combined with the high level of dependence on other barriers, showing that the solution of EV users-related challenges can be facilitated by the solution of other barriers.

The technical challenges related to the V2G technology, show a medium-to-high level of risk, combined with the high level of autonomy from the other barriers. Therefore, the solution to these barriers is important for the adoption of V2G technology, but has to be tackled

separately and in parallel to the others, due to their relatively low influence and dependence scores.

The most notable barriers are related to the business aspects of the adoption of V2G technology, representing significant risks from the one side, and influencing the dependent barriers from the other side. These results show that the solution to these barriers, along with the importance for the adoption of V2G technology, would cause positive chain effects helping to solve the other dependent barriers. This would allow to optimize the efforts of the involved stakeholders to adopt the V2G technology.

The barrier related to the interaction with parking for charging business model is of high importance. Based on the data retrieved from the expert stakeholders of the EV charging business ecosystem, this barrier has the highest risk score, combined with the high level of autonomy from other barriers. This is an important result, showing that the stakeholders perceive high risk in the introduction of the potentially cannibalizing V2G-enabled value propositions in the business models of the currently existing EV charging business ecosystem. Considering the low level of dependence of this barrier, it is unlikely to be solved by solving the other barriers. Therefore, additional research, outlining the opportunities for integration of complimentary non-cannibalizing V2G-related value propositions into the business models of the potential stakeholders, is crucial for the adoption of V2G technology.

Funding

This research was funded by H2020 Research and Innovation Programme, grant number 875131.

CRedit authorship contribution statement

Andrei Goncearuc: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Cedric De Cauwer:** Supervision, Conceptualization, Funding acquisition, Investigation, Writing – original draft, Writing – review & editing. **Nikolaos Sapountzoglou:** Writing – original draft, Visualization, Validation, Supervision, Conceptualization, Formal analysis, Writing – review & editing. **Gilles Van Kriekinghe:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Dominik Huber:** Conceptualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. **Maarten Messagie:** Funding acquisition. **Thierry Coosemans:** Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andrei Goncearuc reports financial support, administrative support, article publishing charges, and writing assistance were provided by VUB University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

Appendix

See Fig. A.1.

Barrier.Nº	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total Influence
1	0	0	0	0	2	2	2	0	2	2	1	1	1	1	2	1	2	3	3	0	3	3	0	31
2	1	0	0	2	1	2	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	1	0	10
3	0	0	0	1	0	2	0	0	1	2	3	0	0	0	1	0	0	0	0	0	0	0	0	10
4	1	0	0	0	1	0	0	0	0	0	1	0	1	0	0	0	1	3	0	0	0	1	0	9
5	0	1	0	0	0	2	0	0	0	0	1	0	0	1	1	0	0	0	0	0	3	3	2	14
6	1	1	0	0	2	0	2	1	0	0	3	0	0	0	1	0	0	0	0	3	1	1	1	17
7	0	1	0	2	3	3	0	3	0	0	2	0	0	1	0	0	0	0	0	2	0	0	0	17
8	0	0	0	1	2	2	3	0	0	0	2	0	0	0	0	0	0	0	0	2	0	1	0	13
9	3	0	1	0	1	0	0	0	0	0	3	0	1	0	0	0	0	1	0	0	0	3	0	13
10	1	0	0	0	1	0	0	0	2	0	3	0	0	0	0	0	0	3	0	0	0	2	1	13
11	0	0	0	0	1	1	0	0	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	6
12	1	0	1	1	0	0	0	0	3	2	0	0	1	0	0	0	0	1	0	0	1	1	0	12
13	2	0	0	2	0	0	0	0	0	1	0	0	0	0	0	0	1	3	2	0	0	2	0	13
14	2	0	2	0	3	2	3	2	0	2	1	0	3	0	0	1	0	2	0	3	0	2	3	31
15	0	0	0	2	0	0	0	0	0	0	0	0	3	0	0	0	0	3	0	0	3	3	2	16
16	1	0	0	1	1	1	0	0	0	1	0	0	2	0	0	0	0	3	0	1	2	2	2	17
17	1	0	1	0	0	0	0	0	1	0	0	0	3	1	0	0	0	2	2	0	0	0	0	11
18	1	0	0	1	0	0	0	0	0	1	0	0	3	0	0	0	3	0	0	0	1	3	2	15
19	0	0	1	0	0	0	0	0	0	0	0	0	3	0	0	0	0	2	0	0	0	0	0	6
20	0	0	1	0	2	0	2	2	0	0	0	0	0	0	0	0	0	1	0	0	0	0	2	10
21	3	0	0	3	1	1	0	0	0	3	0	0	2	0	0	0	0	0	0	1	0	3	2	19
22	1	2	1	1	3	1	0	0	1	2	0	1	2	0	2	0	0	3	0	0	3	0	0	23
23	0	0	1	2	2	2	0	0	3	3	2	0	3	0	0	0	1	2	0	2	3	3	0	29
Total Dependence	19	5	9	19	26	21	12	8	16	20	22	2	28	4	10	2	8	32	7	14	20	34	17	

Fig. A.1. MDI of barriers for V2G adoption.

References

Amsterdam University of Applied Sciences, 0000. A V2G-Repository: 18 European Vehicle2Grid-projects, URL <https://northsearegion.eu/media/4308/v2g-projects-in-europe.pdf>.

Anon, 2017. Commission Regulation (EU) 2017/2195 of 23 November 2017 Establishing a Guideline on Electricity Balancing. European Commission, URL https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=uriserv:OJ.L_.2017.312.01.0006.01.ENG&toc=OJ.L.2017.312:TOC#d1e813-6-1.

Anon, 2019a. Road Vehicles — Vehicle to Grid Communication Interface. International Organization for Standardization.

Anon, 2019b. Interoperability of Public Electric Vehicle Charging Infrastructure. EPRI, URL <https://www.epri.com/research/products/000000003002017164>.

Anon, 2020. Methodology for a List of Standard Products for Balancing Capacity for Frequency Restoration Reserves and Replacement Reserves. European Union Agency for the Cooperation of Energy Regulators (ACER), URL <https://www.acer.europa.eu/en/Electricity/MARKET-CODES/ELECTRICITY-BALANCING/17%20SPBC/Approved/Action%202%20-%20SPBC%20ACER%20decision%20annex%20I.pdf>.

Anon, 2021a. Infrastructure for Charging Electric Vehicles: More Charging Stations but Uneven Deployment Makes Travel Across the EU Complicated. European Court of Auditors, Luxembourg.

Anon, 2021b. RDI Implementation Article 2021–2025. European Network of Transmission System Operators for Electricity, Brussels.

Anon, 2021c. Towards a Consumer-Centric and Sustainable Electricity system. Elia, URL https://issuu.com/eliagroup/docs/20210618_elia_ccmd_white_paper_en?fr=sOTUxNzUyNzcyMTg.

Anon, 2022. Horizon Europe Work Programme 2021–2022. 8. Climate, Energy and Mobility. European Commission, Brussels.

Anon, 2023a. Open Charge Point Protocol 2.0.1. Open Charge Alliance, URL <https://www.openchargealliance.org/protocols/ocpp-201/>.

Anon, 2023b. Mobilize V2G: Where the Future Electric Renault 5 Becomes a Source of Energy. Mobilize, URL <https://media.mobilize.com/mobilize-v2g-where-the-future-electric-reault-5-becomes-a-source-of-energy/?lang=eng>.

Bao, C., Wan, J., Wu, D., Li, J., 2021. Aggregating risk matrices under a normative framework. *J. Risk Res.* 24 (8), 999–1015.

Bartolini, A., Comodi, G., Salvi, D., Østergaard, P.A., 2020. Renewables self-consumption potential in districts with high penetration of electric vehicles. *Energy* 213, 118653.

Baure, G., Dubarry, M., 2020. Durability and reliability of EV batteries under electric utility grid operations: Impact of frequency regulation usage on cell degradation. *Energies* 13 (10), <http://dx.doi.org/10.3390/en13102494>, URL <https://www.mdpi.com/1996-1073/13/10/2494>.

Bishop, J.D., Axon, C.J., Bonilla, D., Tran, M., Banister, D., McCulloch, M.D., 2013. Evaluating the impact of V2G services on the degradation of batteries in PHEV and EV. *Appl. Energy* 111, 206–218. <http://dx.doi.org/10.1016/j.apenergy.2013.04.094>, URL <https://www.sciencedirect.com/science/article/pii/S0306261913004121>.

De Cauwer, C., Van Krieking, G., Van Mierlo, J., Coosemans, T., Messagie, M., 2021. Integration of Vehicle-to-Grid in Local Energy Systems: Concepts and Specific Requirements. *EVS33*, <http://dx.doi.org/10.5281/zenodo.4026726>.

Dubarry, M., Devie, A., McKenzie, K., 2017. Durability and reliability of electric vehicle batteries under electric utility grid operations: Bidirectional charging impact analysis. *J. Power Sources* 358, 39–49. <http://dx.doi.org/10.1016/j.jpowsour.2017.05.015>, URL <https://www.sciencedirect.com/science/article/pii/S0378775317306365>.

Duperrin, J.-C., Godet, M., 1973. Méthode de hiérarchisation des éléments d'un système: essai de prospective du système de l'énergie nucléaire dans son contexte social (Ph.D. thesis). Centre national de l'entrepreneuriat (CNE); CEA.

Elia, 2019. Terms and conditions for balancing service providers for automatic Frequency Restoration Reserve (aFRR). URL https://www.elia.be/-/media/project/elia/elia-site/electricity-market-and-system---document-library/balancing---balancing-services-and-bsp/2020/20200303_tc-bsp-afr-en.pdf.

Elia, 2020a. Terms and Conditions for balancing service providers for Frequency Containment Reserve (FCR). URL <https://www.elia.be/-/media/project/elia/elia-site/public-consultations/2020/20200317-tc-bsp-fcrfinal-consuln.pdf>.

Elia, 2020b. Terms and conditions for balancing service providers for manual Frequency Restoration Reserve (mFRR). URL https://www.elia.be/-/media/project/elia/elia-site/public-consultations/20191004_terms_and_conditions_mfrr/tc-bsp-mfrrfeb2020.pdf.

Elia group, 2020. Accelerating to Net-Zero: Redefining Energy and Mobility. Brussels.

Elia Transmission Belgium, 2021a. Market facilitation. URL <https://innovation.eliagroup.eu/innovation-pillars/market-facilitation/>.

Elia Transmission Belgium, 2021b. Using electric vehicles to balance the network. URL <https://innovation.eliagroup.eu/projects/v2g/>.

Esmaili, M., Shafiee, H., Aghaei, J., 2018. Range anxiety of electric vehicles in energy management of microgrids with controllable loads. *J. Energy Storage* 20, 57–66.

European Council, 2024. Fit for 55, URL <https://www.consilium.europa.eu/en/policies/green-deal/fit-for-55-the-eu-plan-for-a-green-transition/>.

European Council, 2024. Infographic - Fit for 55: towards more sustainable transport, URL <https://www.consilium.europa.eu/en/infographics/fit-for-55-afr-alternative-fuels-infrastructure-regulation/>.

Garvey P.R., L.Z., 1998. Risk matrix: An approach for identifying, assessing, and ranking program risks. *Air Force J. Logist.* 22, 18–21.

Geske, J., Schumann, D., 2018. Willing to participate in vehicle-to-grid (V2G)? Why not! *Energy Policy* 120, 392–401. <http://dx.doi.org/10.1016/j.enpol.2018.05.004>.

Goncearuc, A., Sapountzoglou, N., De Cauwer, C., Coosemans, T., Messagie, M., Crispeels, T., 2022. An integrative approach for business modelling: Application to the EV charging market. *J. Bus. Res.* 143, 184–200.

- Goncearuc, A., Sapountzoglou, N., De Cauwer, C., Coosemans, T., Messagie, M., Crispeels, T., 2023. Incremental profitability evaluation of vehicle-to-grid-enabled automated frequency restoration reserve services for semi-public charging infrastructure: A case study in Belgium. *World Electr. Veh. J.* 14 (12), 339.
- Grajek, M., Kretschmer, T., 2008. Critical Mass. Ludwig Maximilian University of Munich, Munich, URL https://www.researchgate.net/publication/5018482_Critical_Mass.
- Guo, J., Yang, J., Lin, Z., Serrano, C., Cortes, A.M., 2019. Impact analysis of V2G services on EV battery degradation -A review. In: 2019 IEEE Milan PowerTech. pp. 1–6. <http://dx.doi.org/10.1109/PTC.2019.8810982>.
- Haddadian, G., Khodayar, M., Shahidehpour, M., 2015. Accelerating the global adoption of electric vehicles: barriers and drivers. *Electr. J.* 28 (10), 53–68.
- Han, W., Xiao, Y., 2016. Privacy preservation for V2G networks in smart grid: A survey. *Comput. Commun.* 91–92, 17–28. <http://dx.doi.org/10.1016/j.comcom.2016.06.006>.
- Hoang, D.T., Wang, P., Niyato, D., Hossain, E., 2017. Charging and discharging of plug-in electric vehicles (PEVs) in vehicle-to-grid (V2G) systems: A cyber insurance-based model. *IEEE Access* 5, 732–754.
- Hu, H.-Y., Chiu, S.-I., Yen, T.-M., 2009. Modified IPA for order-winner criteria improvement: a MICMAC approach. *J. Appl. Sci.* 9 (21), 3792–3803.
- Huang, B., Meijssen, A.G., Annema, J.A., Lukszo, Z., 2021. Are electric vehicle drivers willing to participate in vehicle-to-grid contracts? A context-dependent stated choice experiment. *Energy Policy* 156, <http://dx.doi.org/10.1016/j.enpol.2021.112410>.
- Huber, D., De Clerck, Q., De Cauwer, C., Sapountzoglou, N., Coosemans, T., Messagie, M., 2021. Vehicle to grid impacts on the total cost of ownership for electric vehicle drivers. *World Electr. Veh. J.* 12 (4), <http://dx.doi.org/10.3390/wevj12040236>.
- Hussain, S.S., Ustun, T.S., Nsonga, P., Ali, I., 2018. IEEE 1609 WAVE and IEC 61850 standard communication based integrated EV charging management in smart grids. *IEEE Trans. Veh. Technol.* 67 (8), 7690–7697.
- IEA, 0000. Global EV outlook 2023, URL <https://www.iea.org/articles/global-ev-outlook-2023>.
- Interreg, 2020. State-of-the-art assessment of smart charging and vehicle 2 grid services. URL <https://www.seev4-city.eu/wp-content/uploads/2020/08/State-of-the-Art-Assessment-of-Smart-Charging-and-V2G-services.pdf>.
- Ioakimidis, C.S., Thomas, D., Rycerski, P., Genikomsakis, K.N., 2018. Peak shaving and valley filling of power consumption profile in non-residential buildings using an electric vehicle parking lot. *Energy* 148, 148–158. <http://dx.doi.org/10.1016/j.energy.2018.01.128>.
- Ismail, A.A., Mbungu, N.T., Elnady, A., Bansal, R.C., Hamid, A.-K., AlShabi, M., 2023. Model of electric vehicles on smart grid and future predictions: A survey. *Int. J. Modelling Simul.* 43 (6), 1041–1057.
- Kempton, W., Tomić, J., 2005. Vehicle-to-grid power fundamentals: Calculating capacity and net revenue. *J. Power Sources* 144 (1), 268–279.
- Kester, J., de Rubens, G.Z., Sovacool, B.K., Noel, L., 2019. Public perceptions of electric vehicles and vehicle-to-grid (V2G): Insights from a Nordic focus group study. *Transp. Res. D* 74, 277–293. <http://dx.doi.org/10.1016/j.trd.2019.08.006>.
- Kester, J., Noel, L., Zarazua de Rubens, G., Sovacool, B.K., 2018. Promoting vehicle to grid (V2G) in the nordic region: Expert advice on policy mechanisms for accelerated diffusion. *Energy Policy* 116, 422–432. <http://dx.doi.org/10.1016/j.enpol.2018.02.024>, URL <https://www.sciencedirect.com/science/article/pii/S0301421518300995>.
- Kuang, Y., Chen, Y., Hu, M., Yang, D., 2017. Influence analysis of driver behavior and building category on economic performance of electric vehicle to grid and building integration. *Appl. Energy* 207, 427–437. <http://dx.doi.org/10.1016/j.apenergy.2017.07.006>.
- Kumar, R.R., Alok, K., 2020. Adoption of electric vehicle: A literature review and prospects for sustainability. *J. Clean. Prod.* 253, 119911.
- Lenka, R.K., Panda, A.K., Dash, A.R., Venkataramana, N.N., Tiwary, N., 2021. Reactive power compensation using vehicle-to-grid enabled bidirectional off-board EV battery charger. In: 2021 1st International Conference on Power Electronics and Energy. ICPEE, IEEE, pp. 1–6.
- Li, S., Gu, C., Zeng, X., Zhao, P., Pei, X., Cheng, S., 2021a. Vehicle-to-grid management for multi-time scale grid power balancing. *Energy* 234, 121201.
- Li, Y., Han, M., Yang, Z., Li, G., 2021b. Coordinating flexible demand response and renewable uncertainties for scheduling of community integrated energy systems with an electric vehicle charging station: A bi-level approach. *IEEE Trans. Sustain. Energy* 12 (4), 2321–2331.
- Li, Y., Messagie, M., Berecibar, M., Hegazy, O., Abdel-Monem, M., Omar, N., Devroey, L., Van Mierlo, J., 2018. The impact of the vehicle-to-grid strategy on lithium-ion battery ageing process. In: 31st International Electric Vehicle Symposium and Exhibition, EVS 2018 and International Electric Vehicle Technology Conference 2018. EVTeC 2018, cited By 1, URL <https://www.scopus.com/inward/record.uri?eid=s2-s2.0-85073110497&partnerID=40&md5=2419637e6d196e83161ab787fa08f49>.
- Licea, M., 2020. Fault tolerant boost converter with multiple serial inputs and output voltage regulation for vehicle-to-grid services. *Energies* 13 (7), <http://dx.doi.org/10.3390/en13071694>, cited By 2.
- Lu, M.Z., Guo, Z.W., Liaw, C.M., 2021. A battery/supercapacitor hybrid powered EV SRM drive and microgrid incorporated operations. *IEEE Trans. Transp. Electr.* 1, <http://dx.doi.org/10.1109/TTE.2021.3081103>.
- Ma, Y., Houghton, T., Cruden, A., Infield, D., 2012. Modeling the benefits of vehicle-to-grid technology to a power system. *IEEE Trans. Power Syst.* 27 (2), 1012–1020.
- Mojumder, M.R.H., Ahmed Antara, F., Hasanuzzaman, M., Alamri, B., Alsharef, M., 2022. Electric vehicle-to-grid (V2G) technologies: Impact on the power grid and battery. *Sustainability* 14 (21), 13856.
- Mukherjee, B., Kariniotakis, G., Sossan, F., 2021. Smart charging, vehicle-to-grid, and reactive power support from electric vehicles in distribution grids: A performance comparison. In: 2021 IEEE PES Innovative Smart Grid Technologies Europe. ISGT Europe, IEEE, pp. 1–6.
- Mwasilu, F., Justo, J.J., Kim, E.-K., Do, T.D., Jung, J.-W., 2014. Electric vehicles and smart grid interaction: A review on vehicle to grid and renewable energy sources integration. *Renew. Sustain. Energy Rev.* 34, 501–516.
- Noel, L., de Rubens, G.Z., Kester, J., Sovacool, B.K., 2019a. Navigating expert skepticism and consumer distrust: Rethinking the barriers to vehicle-to-grid (V2G) in the Nordic region. *Transp. Policy* 76, 67–77. <http://dx.doi.org/10.1016/j.tranpol.2019.02.002>.
- Noel, L., de Rubens, G., Kester, J., Sovacool, B., 2019b. Vehicle-to-grid: A sociotechnical transition beyond electric mobility. In: *Energy, Climate and the Environment*, Springer International Publishing.
- Noel, L., Zarazua de Rubens, G., Kester, J., Sovacool, B.K., Noel, L., Zarazua de Rubens, G., Kester, J., Sovacool, B.K., 2019c. The technical challenges to V2G. *Vehicle-to-Grid: A Sociotechnical Transition beyond Electric Mobility*. Springer, pp. 65–89.
- Pearre, N.S., Ribberink, H., 2019. Review of research on V2X technologies, strategies, and operations. *Renew. Sustain. Energy Rev.* 105, 61–70. <http://dx.doi.org/10.1016/J.RSER.2019.01.047>.
- PwC, 2019. Making the most of Europe's grids. URL https://www.renaultgroup.com/wp-content/uploads/2020/05/deliverable-2_190517-elaad-final-article-regulatory-barriers-smart-charging-stc.pdf.
- Rahman, M., Hossain, M., Lu, J., Rafi, F., Mishra, S., 2020. A vehicle-to-microgrid framework with optimization-incorporated distributed EV coordination for a commercial neighborhood. *IEEE Trans. Ind. Inform.* 16 (3), 1788–1798. <http://dx.doi.org/10.1109/TII.2019.2924707>, cited By 2.
- Rezvani, Z., Jansson, J., Bodin, J., 2015. Advances in consumer electric vehicle adoption research: A review and research agenda. *Transp. Res. D* 34, 122–136. <http://dx.doi.org/10.1016/j.trd.2014.10.010>.
- Rodríguez-Licea, M.-A., Perez-Pinal, F.-J., Soriano-Sánchez, A.-G., Vázquez-López, J.-A., 2019. Noninvasive vehicle-to-load energy management strategy to prevent li-ion batteries premature degradation. *Math. Probl. Eng.* 2019, <http://dx.doi.org/10.1155/2019/8430685>, cited By 2.
- Saldana, G., Martin, J.I.S., Zamora, I., Asensio, F.J., Oñederra, O., 2019. Electric vehicle into the grid: Charging methodologies aimed at providing ancillary services considering battery degradation. *Energies* 12 (12), <http://dx.doi.org/10.3390/en12122443>.
- Salvati, G.A., Carati, E.G., Cardoso, R., da Costa, J.P., Stein, C.M.d., 2020. Electric vehicles energy management with V2G/G2V multifactor optimization of smart grids. *Energies* 13 (5), 1191.
- Saxena, N., Grijalva, S., Chukwuka, V., Vasilakos, A.V., 2017. Network security and privacy challenges in smart vehicle-to-grid. *IEEE Wirel. Commun.* 24, 88–98. <http://dx.doi.org/10.1109/MWC.2016.1600039WC>.
- Scaringella, L., Radziwon, A., 2018. Innovation, entrepreneurial, knowledge, and business ecosystems: Old wine in new bottles? *Technol. Forecast. Soc. Change* 136, 59–87.
- Sovacool, B.K., Axsen, J., Kempton, W., 2017. The future promise of vehicle-to-grid (V2G) integration: a sociotechnical review and research agenda. *Annu. Rev. Environ. Resour.* 42, 377–406.
- Steward, D.M., 2017. Critical Elements of Vehicle-To-Grid (V2g) Economics. National Renewable Energy Lab.(NREL), Golden, CO (United States).
- Tan, K.M., Padmanaban, S., Yong, J.Y., Ramachandramurthy, V.K., 2019. A multi-control vehicle-to-grid charger with bi-directional active and reactive power capabilities for power grid support. *Energy* 171, 1150–1163.
- Thingvad, A., Calearo, L., Andersen, P., Marinelli, M., 2021. Empirical capacity measurements of electric vehicles subject to battery degradation from V2G service. *IEEE Trans. Veh. Technol.* <http://dx.doi.org/10.1109/TVT.2021.3093161>.
- Thingvad, A., Marinelli, M., 2019. Influence of V2G frequency services and driving on electric vehicles battery degradation in the nordic countries. In: *Proceedings of EVS 31 & EVTeC 2018. 31st International Electric Vehicles Symposium ; Exhibition; International Electric Vehicle Technology Conference 2018, EVS31 and EVTeC2018 ; Conference date: 30-09-2018 Through 03-10-2018*.
- Thompson, A.W., Perez, Y., 2020. Vehicle-to-Everything (V2X) energy services, value streams, and regulatory policy implications. *Energy Policy* 137 (December 2019), 111136. <http://dx.doi.org/10.1016/j.enpol.2019.111136>.
- Uddin, K., Dubarry, M., Glick, M.B., 2018. The viability of vehicle-to-grid operations from a battery technology and policy perspective. *Energy Policy* 113, 342–347. <http://dx.doi.org/10.1016/j.enpol.2017.11.015>, URL <https://www.sciencedirect.com/science/article/pii/S0301421517307619>.

- van Heuveln, K., Ghotge, R., Annema, J.A., van Bergen, E., van Wee, B., Pesch, U., 2021. Factors influencing consumer acceptance of vehicle-to-grid by electric vehicle drivers in the Netherlands. *Travel Behav. Soc.* 24, 34–45. <http://dx.doi.org/10.1016/j.tbs.2020.12.008>, URL <https://www.sciencedirect.com/science/article/pii/S2214367X20302519>.
- Van Kriekinge, G., De Cauwer, C., Sapountzoglou, N., Coosemans, T., Messagie, M., 2021. Peak shaving and cost minimization using model predictive control for uni- and bi-directional charging of electric vehicles. *Energy Artic.* 7, 8760–8771. <http://dx.doi.org/10.1016/j.egy.2021.11.207>.
- Wang, D., Coignard, J., Zeng, T., Zhang, C., Saxena, S., 2016. Quantifying electric vehicle battery degradation from driving vs. vehicle-to-grid services. *J. Power Sources* 332, 193–203. <http://dx.doi.org/10.1016/j.jpowsour.2016.09.116>, URL <https://www.sciencedirect.com/science/article/pii/S0378775316313052>.
- Wang, Y., Sheikh, O., Hu, B., Chu, C.-C., Gadh, R., 2014. Integration of V2H/V2G hybrid system for demand response in distribution network. In: 2014 IEEE International Conference on Smart Grid Communications. SmartGridComm, pp. 812–817. <http://dx.doi.org/10.1109/SmartGridComm.2014.7007748>.
- Wang, Z., Wang, S., 2013. Grid power peak shaving and valley filling using vehicle-to-grid systems. *IEEE Trans. Power Deliv.* 28 (3), 1822–1829.
- WindEurope, 2020. Making the most of Europe's grids. URL <https://windeurope.org/wp-content/uploads/files/policy/position-papers/20200922-WindEurope-Grid-optimisation-technologies-to-build-a-greener-Europe.pdf>.
- Wu, G., Zhang, X., Dong, Z., 2015. Powertrain architectures of electrified vehicles: Review, classification and comparison. *J. Franklin Inst.* 352 (2), 425–448.
- Yamagata, Y., Seya, H., Kuroda, S., 2014. Energy resilient smart community: Sharing green electricity using V2C technology. *Energy Procedia* 61, 84–87. <http://dx.doi.org/10.1016/j.egypro.2014.11.912>, cited By 5.
- Yi, Z., Scofield, D., Smart, J., Meintz, A., Jun, M., Mohanpurkar, M., Medam, A., 2020. A highly efficient control framework for centralized residential charging coordination of large electric vehicle populations. *Int. J. Electr. Power Energy Syst.* 117 (June 2019), 105661. <http://dx.doi.org/10.1016/j.ijepes.2019.105661>.
- Yilmaz, M., Krein, P.T., 2012. Review of benefits and challenges of vehicle-to-grid technology. In: 2012 IEEE Energy Conversion Congress and Exposition. ECCE, IEEE, pp. 3082–3089.
- Yu, H., Niu, S., Shang, Y., Shao, Z., Jia, Y., Jian, L., 2022. Electric vehicles integration and vehicle-to-grid operation in active distribution grids: A comprehensive review on power architectures, grid connection standards and typical applications. *Renew. Sustain. Energy Rev.* 168, 112812.
- Zhang, J., Fan, Y., 2010. Current state and research trends on business ecosystem. In: 2010 IEEE International Conference on Service-Oriented Computing and Applications. SOCA, IEEE, pp. 1–5.